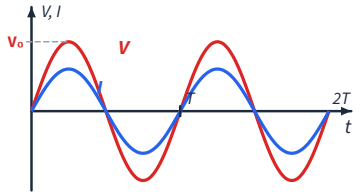
# CHAPTER 18 ALTERNATING CURRENT

**Key idea:** the current and voltage vary sinusoidally and reverse direction every half cycle.

## 1 BASIC EQUATIONS




Pure resistor: $V$ and $I$ in phase

$$I = I_0 \sin \omega t \qquad V = V_0 \sin \omega t$$

$I_0, V_0$ peak values

$\omega = 2\pi f$ (rad s$^{-1}$)

$T = \frac{1}{f}$ (s)

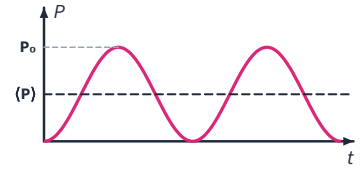
## 2 R.M.S. (EFFECTIVE) VALUE

$$I_{\text{rms}} = \frac{I_0}{\sqrt{2}} \qquad V_{\text{rms}} = \frac{V_0}{\sqrt{2}}$$

- The steady d.c. that dissipates heat in a resistor **at the same rate** as the a.c.
- Mean value over one cycle $= 0$
- Mains 240 V is r.m.s.: $V_0 = 339$ V

## 3 POWER IN A.C.

$$\langle P \rangle = I_{\text{rms}} V_{\text{rms}} = I_{\text{rms}}^2 R = \tfrac{1}{2} I_0 V_0$$




- Peak power $P_0 = I_0^2 R = 2\langle P \rangle$
- Instantaneous $P$ varies at **twice** the supply frequency, never negative

Pure $L$ and pure $C$: mean power $= 0$. Energy is stored, then returned to the source.

## 4 COMPONENTS

### RESISTOR $R$

- $V$ and $I$ **in phase**, $\phi = 0$
- $R$ independent of $f$
- Dissipates energy as heat

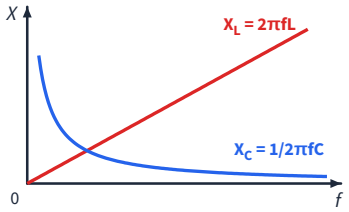
### INDUCTOR $L$

- $X_L = \omega L = 2\pi f L$
- $V$ **leads** $I$ by 90°
- d.c.: $X_L = 0$

### CAPACITOR $C$

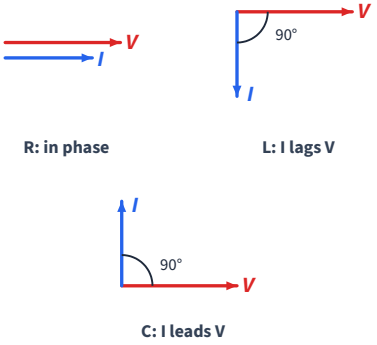
- $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$
- $I$ **leads** $V$ by 90°
- Blocks d.c.: $X_C \to \infty$

## 5 REACTANCE AND FREQUENCY




- $X_L \propto f$: straight line through origin, gradient $2\pi L$
- $X_C \propto \frac{1}{f}$: falls towards zero, never touches the axis
- $f$ doubled: $X_L$ doubles, $X_C$ halves

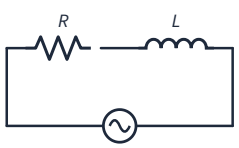
## 6 PHASE (PHASORS)




Series R–L or R–C: phase angle $0 < \phi < 90^\circ$

**ELI the ICE man:** in L, e.m.f. leads I; in C, I leads e.m.f.

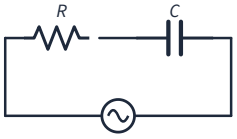
## 7 R-L AND R-C SERIES CIRCUITS




$$Z = \sqrt{R^2 + X_L^2}$$

$$\tan \phi = \frac{X_L}{R}$$

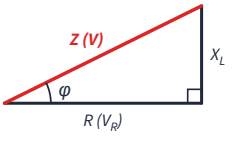
**R–L:** current **lags** voltage by $\phi$




$$Z = \sqrt{R^2 + X_C^2}$$

$$\tan \phi = \frac{X_C}{R}$$

**R–C:** current **leads** voltage by $\phi$




$$V^2 = V_R^2 + V_L^2$$

$$\cos \phi = \frac{R}{Z}$$

$V_{\text{rms}} = I_{\text{rms}} Z$ and mean power $\langle P \rangle = I_{\text{rms}} V_{\text{rms}} \cos \phi = I_{\text{rms}}^2 R$ (only $R$ dissipates power).

## 8 UNIT CHECK

| Quantity | Formula | Unit |
|---|---|---|
| Period | $T = 1/f$ | s |
| Angular frequency | $\omega = 2\pi f$ | rad s$^{-1}$ |
| R.m.s. current | $I_{\text{rms}} = I_0/\sqrt{2}$ | A |
| Inductive reactance | $X_L = \omega L$ | Ω |
| Capacitive reactance | $X_C = 1/\omega C$ | Ω |
| Impedance | $Z = \sqrt{R^2 + X^2}$ | Ω |
| Power factor | $\cos \phi = R/Z$ | no unit |
| Mean power | $I_{\text{rms}}^2 R$ | W |

## 9 KEY POINTS TO MEMORISE

1. R.m.s. value = peak value $\div \sqrt{2}$ (sine wave only).
2. Only the resistor dissipates mean power.
3. Inductor stores energy in its **magnetic** field; capacitor in its **electric** field.
4. Voltages across $R$ and $X$ are 90° apart: add as **phasors**, not arithmetically.

## 10 COMMON SECTION A TRAPS

- Reading $\omega$ (314 rad s$^{-1}$) as the frequency: $f = \omega/2\pi = 50$ Hz.
- Using $V_0$ in $P = V^2/R$ gives the **peak** power, twice the mean.
- Forgetting μF → ×10$^{-6}$ F or mH → ×10$^{-3}$ H.
- Stating that current leads voltage in an inductor (that is the capacitor).

## EXAM TIPS

- ✓ Convert peak values to r.m.s. before finding power.
- ✓ Find $X_L$ or $X_C$ first, then $Z$, then $I$.
- ✓ Draw the phasor (impedance) triangle: $R$ across, $X$ up, $Z$ hypotenuse.
- ✓ State the phase relation in words: "current leads/lags voltage by …".